

# Analysis Pipeline for:

## Genome-Wide Analysis Reveals Altitude-Associated Divergence in a Color-Polymorphic Insect

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# 1 Generation of Master BEAGLE File

The BEAGLE file contains normalized genotype likelihoods used in down-stream PCA and Admixture analyses.

```
angsd -bam isophya71.bamlist \
  -ref isophya_contigs_CAYMY.fasta \
  -out isophya71 \
  -GL 1 \
  -doMajorMinor 1 \
  -doMaf 1 \
  -doGlf 2 \
  -SNP_pval 1e-12 \
  -minMaf 0.05 \
  -sites 6cov_nonparalog.sites
```

## 2 Principal Component Analysis (PCA)

### Step 1: Estimation of the Genetic Covariance Matrix

```
pcangsd -b isophya71.beagle.gz --snp_weight -o isophya71
```

### Step 2: Eigen Decomposition and Plotting

```
Rscript plotPCA.R -i isophya71.cov -c 1-2 -a isophya71.clst -o isophya71_pca_1_2.pdf
```

### Step 3: QQ Plots for SNP Loadings on PC1 and PC2

```
infile=isophya71.pca.loadings
Rscript plotQQ.R
```

## 3 Admixture Analysis

### Step 1: Calculate Admixture Proportions

Calculate proportions for  $K = 1$  to 5 with 10 independent replicates per  $K$ .

```
K={1..5}
RUN={1..10}
NGSadmix -likes isophya71.beagle.gz \
  -K ${K} \
  -seed ${RANDOM} \
  -o isophya71_admix_${K}_run_${RUN}
```

## Step 2: Cross-Validation via Evanno's Method

Choose the optimal  $K$  value and generate a  $\Delta K$  plot.

```
infile=isophya71.admix.runs.LH.txt
Rscript Evanno_DeltaK.R
```

## Step 3: Visualizing Admixture Plots ( $K = 2-5$ )

```
infile={isophya71.info; isophya71.admix[2-5].run1.qopt}
Rscript plotAdmix.R
```

# 4 Pairwise Fixation Index ( $F_{ST}$ ) and IBD

## Step 1: Generate Folded SFS for All Collection Sites

```
pop={450, 850, 900, 1000, 1100, 1200, 1300, 1900, 2000, 2100, 2300}

## Compute Sample Allele Frequencies (SAF)
angsd -bam ${pop}.bamlist \
  -ref isophya_contigs_CAYMY.fasta \
  -anc isophya_contigs_CAYMY.fasta \
  -out ${pop} \
  -GL 1 \
  -doSaf 1 \
  -doCounts 1 \
  -sites 6cov_nonparalog.sites

## Optimize Site Frequency Spectrum via realSFS
realSFS ${pop}.saf.idx -maxIter 100 -fold 1 > ${pop}.folded.sfs
```

## Step 2: Plot SFS

Infiles: \*.folded.sfs

```
infile=*.folded.sfs
Rscript plotSFS.R
```

## Step 3: Calculate Pairwise $F_{ST}$ Values

```
pop={450, 850, 900, 1000, 1100, 1200, 1300, 1900, 2000, 2100, 2300}

## Estimate the Folded 2D-SFS from SAF data (-fold 1)
realSFS ${SAF1} ${SAF2} -maxIter 100 -fold 1 -P 2 > ${pop1}.${pop2}.folded.2dsfs

## Build the pairwise Fst alignment index map (-fold 1)
realSFS fst index ${SAF1} ${SAF2} -fold 1 -whichFst 1 \
  -sfs ${pop1}.${pop2}.folded.2dsfs \
  -fstout ${pop1}.${pop2}.folded
```

```
## Extract global weighted Fst estimates
realSFS fst stats ${pop1}.${pop2}.folded.fst.idx 2> ${pop1}.${pop2}.folded.global.fst
```

## Step 4: Run Mantel Test and Plot IBD

```
infile=isophya71.fst.dist.tsv
Rscript run_mantel_test.R
Rscript plotIBD.R
Rscript plotFst.R
```

# 5 Partitioning Neutral and Putatively Adaptive Loci

## Step 1: Calculate HWE Outliers Due to Inbreeding

```
pcangsd -b isophya71.beagle.gz \
-o isophya71 \
--inbreed_samples \
--inbreed_sites
```

## Step 2: Determine Putatively Adaptive Loci via pcadapt

```
pcangsd -b isophya71.beagle.gz \
--hwe isophya71.lrt.sites \
--pcadapt \
--sites_save \
-o isophya71
```

# 6 Comparing Neutral and Adaptive Genetic Variation

## Step 1: Generate SFS for Putatively Adaptive Loci

```
pop={450, 850, 900, 1000, 1100, 1200, 1300, 1900, 2000, 2100, 2300}

## Compute Sample Allele Frequencies (SAF)
angsd -bam ${pop}.bamlist \
-ref isophya_contigs_CAYMY.fasta \
-anc isophya_contigs_CAYMY.fasta \
-out ${pop}.adapt \
-GL 1 \
-doSaf 1 \
-doCounts 1 \
-sites 6cov_nonparalog.adapt.sites

## Optimize Site Frequency Spectrum via realSFS
realSFS ${pop}.adapt.saf.idx -maxIter 100 -fold 1 > ${pop}.adapt.folded.sfs
```

## Step 2: Calculate Adaptive Theta Statistics

```
pop={450, 850, 900, 1000, 1100, 1200, 1300, 1900, 2000, 2100, 2300}
realSFS saf2theta ${pop}.adapt.saf.idx -sfs ${pop}.adapt.sfs ${pop}.adapt
thetaStat do_stat ${pop}.adapt.thetas.idx
```

## Step 3: Calculate Pairwise $F_{ST}$ for Adaptive Loci

```
pop={450, 850, 900, 1000, 1100, 1200, 1300, 1900, 2000, 2100, 2300}
realSFS ${SAF1} ${SAF2} -maxIter 100 -fold 1 -P 2 > ${pop1}.${pop2}.adapt.folded.2dsfs

realSFS fst index ${SAF1} ${SAF2} -fold 1 -whichFst 1 \
  -sfs ${pop1}.${pop2}.adapt.folded.2dsfs \
  -fstout ${pop1}.${pop2}.adapt.folded

realSFS fst stats ${pop1}.${pop2}.adapt.folded.fst.idx 2>
  ${pop1}.${pop2}.adapt.folded.global.fst
```

## Step 4: Generate SFS for Neutral Loci

```
pop={450, 850, 900, 1000, 1100, 1200, 1300, 1900, 2000, 2100, 2300}

## Compute Sample Allele Frequencies (SAF)
angsd -bam ${pop}.bamlist \
  -ref isophya_contigs_CAYMY.fasta \
  -anc isophya_contigs_CAYMY.fasta \
  -out ${neut}.adapt \
  -GL 1 \
  -doSaf 1 \
  -doCounts 1 \
  -sites 6cov_nonparalog.neut.sites

## Optimize Site Frequency Spectrum via realSFS
realSFS ${pop}.neut.saf.idx -maxIter 100 -fold 1 > ${pop}.neut.folded.sfs
```

## Step 5: Calculate Neutral Theta Statistics

```
pop={450, 850, 900, 1000, 1100, 1200, 1300, 1900, 2000, 2100, 2300}
realSFS saf2theta ${pop}.neut.saf.idx -sfs ${pop}.neut.sfs ${pop}.neut
thetaStat do_stat ${pop}.neut.thetas.idx
```

## Step 6: Calculate Pairwise $F_{ST}$ for Neutral Loci

```
pop={450, 850, 900, 1000, 1100, 1200, 1300, 1900, 2000, 2100, 2300}
realSFS ${SAF1} ${SAF2} -maxIter 100 -fold 1 -P 2 > ${pop1}.${pop2}.neut.folded.2dsfs

realSFS fst index ${SAF1} ${SAF2} -fold 1 -whichFst 1 \
  -sfs ${pop1}.${pop2}.neut.folded.2dsfs \
```

```
-fstout ${pop1}.${pop2}.neut.folded
realSFS fst stats ${pop1}.${pop2}.neut.folded.fst.idx 2>
    ${pop1}.${pop2}.neut.folded.global.fst
```

## Step 7: Plot Diversity Comparisons

```
infile=isophya71.diversity.tsv
Rscript plotDiversity.R
```

## Step 8: Plot Pairwise $F_{ST}$ Comparisons

```
infile={isophya71.neut.fst.tsv; isophya71.adapt.fst.tsv}
Rscript plotFst.R
```

## Step 9: Plot IBD Comparisons

```
infile={isophya71.neut.fst.tsv; isophya71.adapt.fst.tsv}
Rscript plotIBDmulti.R
```

# 7 Altitude Association Test

```
angsd -bam isophya71.bamlist \
    -ref isophya_contigs_CAYMY.fasta \
    -yQuant altitude.yquant \
    -doAsso 2 \
    -GL 1 \
    -doPost 1 \
    -out isophya71 \
    -doMajorMinor 1 \
    -doMaf 1 \
    -SNP_pval 1e-12 \
    -postCutoff 0.95 \
    -minMaf 0.05 \
    -minHigh 10 \
    -sites 6cov_nonparalog.sites
```

# 8 Allele-Frequency Plots for Altitude Associated Loci

## Step 1: Determine Minor Allele Across All 71 Samples

```
angsd -bam isophya71.bamlist \
    -ref isophya_contigs_CAYMY.fasta \
    -out isophya71.assoc \
    -GL 1 \
    -doMajorMinor 1 \
```

```
-doCounts 1 \
-doMaf 1 \
-sites isophya71.assoc.alt.sites
```

## Step 2: Determine Reference Allele Frequencies per Altitude

```
pop={450, 850, 900, 1000, 1100, 1200, 1300, 1900, 2000, 2100, 2300}
angsd -bam ${pop}.bamlist \
-ref isophya_contigs_CAYMY.fasta \
-out ${pop} \
-GL 1 \
-doMajorMinor 3 \
-doCounts 1 \
-doMaf 1 \
-sites isophya71.assoc.alt.sites.ref
```

## Step 3: Plot Allele Frequency Trends

```
infile=isophya.alt.assoc.loci.mafs.tsv
Rscript plotMafTrend.R
```

# 9 Estimating Cline Parameters of Altitude Associated Loci

## Step 1: Call Genotypes for Altitude-Associated Loci

```
angsd -bam isophya71.bamlist \
-ref isophya_contigs_CAYMY.fasta \
-out isophya71.assoc \
-GL 1 \
-doMajorMinor 1 \
-doMaf 1 \
-doGeno 5 \
-only_proper_pairs 1 \
-doPost 1 \
-postCutoff 0.95 \
-SNP_pval 1e-12 \
-minMaf 0.05 \
-sites isophya71.assoc.alt.sites
```

## Step 2: Calculate Allele Counts and Fit Clines via HZAR

```
infile={isophya71.assoc.genotype.tsv; isophya71.info}
Rscript cline_fitting_hzar_parboot.R
```



### Step 3: Summarize fitted clines

```
infile=hzar_ML_summary.tsv  
Rscript summarize_clines.R
```

## 10 Genetic Discrimination Between Colour Morphs

### Step 1: Call Genotypes

```
angsd -bam isophya71.bamlist \  
-ref isophya_contigs_CAYMY.fasta \  
-out isophya71 \  
-GL 1 \  
-doMajorMinor 1 \  
-doMaf 1 \  
-doBcf 1 \  
-only_proper_pairs 1 \  
-doPost 1 \  
-postCutoff 0.95 \  
-SNP_pval 1e-12 \  
-minMaf 0.05 \  
-sites 6cov_nonparalog.sites \  

```

### Step 2: Run DAPC Analysis and Produce Manhattan Plot

```
infile=isophya71.vcf  
Rscript run_dapc.R
```

### Step 3: Plot Discriminant Function Density Plot

```
infile=isophya71.dapc.df1.csv  
Rscript plotDensity.R
```

### Step 4: Plot Posterior Assignment Probabilities

```
infile=isophya71.dapc.assignment.tsv  
Rscript plotAssignmentDapc.R
```

### Step 5: Plot Genotype Distribution of Major Effect Loci

```
infile=isophya71.dapc.top.snps.tsv  
Rscript plotHeatMapGeno.R
```